GAD vs Sella vs Hybrid GAD–Newton

Aggregated TS-finder benchmark on Transition1x test split

2026-05-16. $n = 287$ samples per (method, noise). Headline + starting-condition + convergence-criterion sensitivity.

TL;DR

Three families, six noise levels (10–200 pm), one starting condition (noised true T1x TS) for the headline sweep, plus a reactant-start probe at 0 pm. Best of each family:

> **Sella cartesian Eckart untuned Hess.Freq.=1** wins TS conv ≤50 pm (92.7% @ 10 pm). **Plain GAD dt=0.005** wins IRC TOPO at ≥100 pm (+11.9 pp over Sella at 150 pm; +21.3 pp at 200 pm). **Hybrid damped Eckart eig-switch tr=0.05** wins *wall-time per converged TS* at every noise (1.4–3.0× vs Sella; 4.1–4.5× vs plain GAD), and produces *geometrically tighter* TSs than either baseline at high noise (p95 RMSD-to-known-TS at 200 pm: hybrid 0.11 Å vs Sella 0.84 Å vs plain GAD 0.46 Å). **Starting from reactant @ 0 pm**, Newton methods dominate (Sella libdef 80.8%; plain GAD partial 40–54%) — GAD's slow walk struggles without a saddle direction in hand.

Sella naming convention (used throughout). The project's historical names (`libdef`, `default`, `internal`) were inconsistent. This report uses a three-axis name:

Sella {cartesian — internal} {Eckart}? {tuned — untuned} Hess.Freq.=N

- **coords**: `cartesian` or `internal` (only Cartesian needs Eckart projection)
- **Eckart**: present only if Eckart projection is applied
- **tuned**: trust-radius hyperparameters (δ_0, γ) tuned from Sella's library defaults. `tuned`: δ_0=0.048, γ=0 (this project's choice); `untuned`: δ_0=0.1, γ=0.4 (Sella library defaults)
- **Hess.Freq.=N**: Hessian recomputed every N steps. Sella library default is 3; our project canonical is 1.

Historical → new name map

Old name	New name (this report)
`Sella libdef` (cart+Eckart, δ_0=0.1, γ=0.4, d=1)	Sella cartesian Eckart untuned Hess.Freq.=1
`Sella default` (cart no-Eckart, δ_0=0.048, γ=0, d=1)	Sella cartesian tuned Hess.Freq.=1
`Sella internal` (internal coords, δ_0=0.048, γ=0, d=1)	Sella internal tuned Hess.Freq.=1
`Sella libdef Hess-freq d=3`	Sella cartesian Eckart untuned Hess.Freq.=3

Recommendation.

Goal	Use	Why
Fastest TS finder, any noise	Hybrid damped Eckart eig-switch tr=0.05	1.4–3× vs Sella
Highest IRC TOPO at $\geq$100 pm	Plain GAD dt=0.005	Right-basin failures
Tightest geometry at high noise	Hybrid damped Eckart eig-switch tr=0.05	7.7× tighter p95
Low-noise TS conv ($\leq$50 pm)	Sella cartesian Eckart untuned Hess.Freq.=1	92.7% @ 10 pm
Starting from reactant @ 0 pm	Sella cartesian Eckart untuned Hess.Freq.=1	80.8% — Newton finds nearby saddle

1 Headline figure 1 — best-of-family across TS noise (noised true TS)

The starting condition for this sweep is the **noised true T1x TS** (Gaussian perturbation of the labelled saddle geometry). Six noise levels, identical sample set, project-canonical convergence criterion $n_{\text{neg}}=1 \wedge F_{\text{max}} < 0.01$ eV/Å.

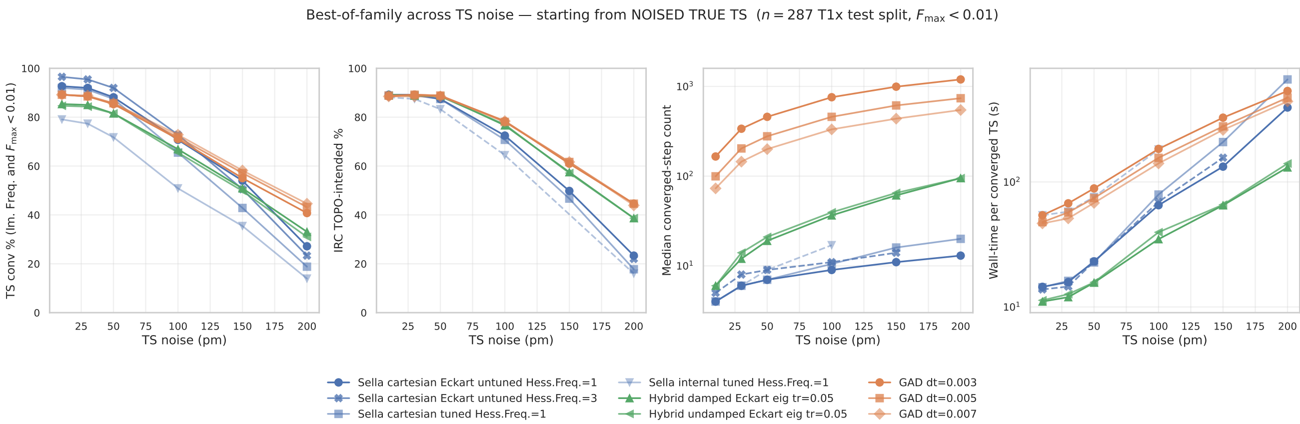

Figure 1: Best-of-family, four diagnostic axes. TS conv, IRC TOPO, median converged-step count, wall-time per converged TS, all as a function of noise level. Each family contributes its strongest configs as separate lines. The Hess.Freq.=3 variant is now available across 10–200 pm (200 pm from pooled main + safety-net partitions, $n = 53$); Sella internal goes 10–100 pm only. All others span the full 10–200 pm grid. Source: `master_2026_05_11.csv`.

Per-noise tables. All numbers from `analysis_2026_04_29/master_2026_05_11.csv`. Bold = best in row. ^{p} = partial: extracted from SLURM log after timeout (biased toward smaller molecules; see Section **??**). Subscripts on ^{p} indicate the sample count when known and the source is pooled (main + safety-net partitions), so the bias is reduced relative to log-only extraction.

Table 1 — TS conv % (Im. Freq. and $F_{\max} < 0.01$, $n = 287$)

method	**10 pm**	**30 pm**	**50 pm**	**100 pm**	**150 pm**	**200 pm**
GAD dt=0.003	89.2	88.5	85.4	71.1	55.1	40.8
GAD dt=0.005	89.2	88.5	85.7	71.8	57.1	43.2
GAD dt=0.007	89.2	88.9	85.7	72.8	58.2	**44.6**
Sella cartesian tuned Hess.Freq.=1	92.0	91.3	87.5	65.5	42.9	18.8
Sella cartesian Eckart untuned Hess.Freq.=1	92.7	92.0	88.2	70.7	54.0	27.2
Sella internal tuned Hess.Freq.=1	79.1	77.4	71.8	50.9	$35.5^{p_{172}}$	13.9
Sella cartesian Eckart untuned Hess.Freq.=3	**96.5**	**95.5**	**92.0**	72.8	50.5	23.3
Hybrid damped Eckart eig tr=0.05	85.4	85.0	81.5	66.9	50.9	33.1
Hybrid undamped Eckart eig tr=0.05	84.7	84.3	81.5	65.5	49.8	31.0

Table 2 — IRC TOPO-intended % (forward+backward Sella IRC + graph-match to T1x reactant/product, $n = 287$)

method	**10 pm**	**30 pm**	**50 pm**	**100 pm**	**150 pm**	**200 pm**
GAD dt=0.003	88.5	**89.2**	88.9	78.4	61.0	44.6
GAD dt=0.005	88.9	**89.2**	88.9	78.0	61.7	**44.6**
GAD dt=0.007	88.5	88.5	88.5	**78.4**	**61.7**	43.9
Sella cartesian tuned Hess.Freq.=1	88.9	88.9	87.5	70.7	46.7	17.8
Sella cartesian Eckart untuned Hess.Freq.=1	**89.2**	**89.2**	87.5	72.5	49.8	23.3
Sella internal tuned Hess.Freq.=1	88.2	87.5	83.3	64.5	—	16.0
Sella cartesian Eckart untuned Hess.Freq.=3	88.5	88.9	87.1	70.4	45.3	22.0
Hybrid damped Eckart eig tr=0.05	**89.2**	88.9	88.9	76.7	57.5	38.7
Hybrid undamped Eckart eig tr=0.05	88.9	88.9	88.5	77.0	57.1	38.7

2 Starting condition: reactant @ 0 pm noise

To probe whether the headline result depends on the noised-TS starting condition, we ran a subset of methods starting from the *reactant* geometry at 0 pm noise — i.e. no perturbation, but a much larger geometric displacement from the saddle than any noise level above. Newton methods dominate here because they navigate to the nearest saddle along a curvature direction; GAD walks slowly and runs out of step budget.

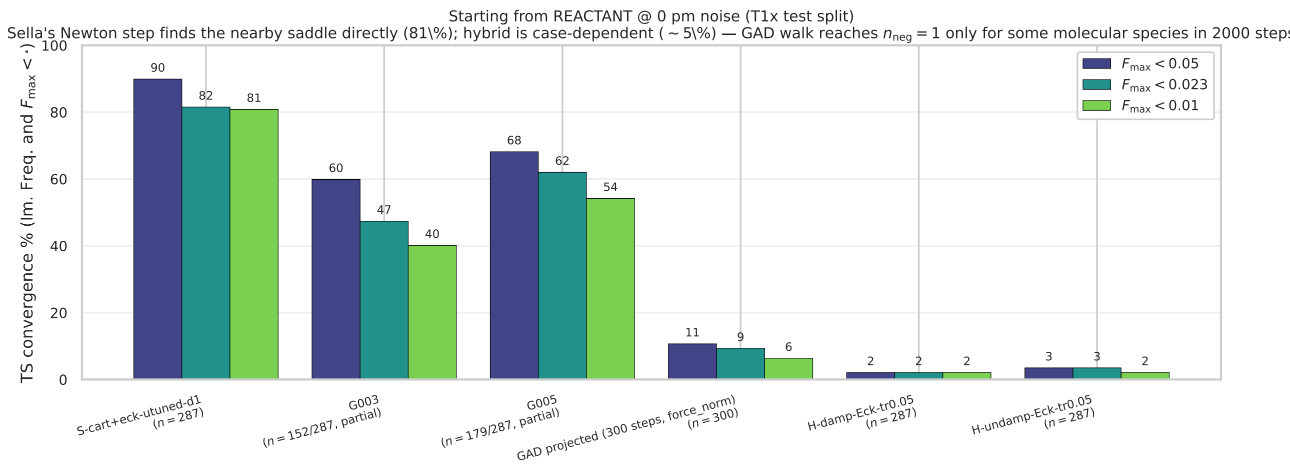

Figure 2: Starting from REACTANT @ 0 pm noise. TS conv % at three $F_{\max}$ thresholds (0.05, 0.023 [Gaussian], 0.01 [project canonical]). **Sella** (cart+Eckart untuned d=1) lands ~81% of saddles even starting from the reactant — Newton step climbs to the closest saddle direction within ~20 iterations. **Plain GAD** (dt=0.003, dt=0.005) needs >1000 steps to walk reactant→saddle and partial-run statistics ($n = 152/287, 179/287$) show 40–54% conv at the project threshold. The 300-step GAD-projected cell (force-norm criterion, no fmax in the trajectory schema) illustrates the step-budget collapse: only 6% converge with a 300-step cap. **Hybrid damped/undamped from reactant: 2.1% / 2.1% conv ($n = 287$, FINAL)** — geometrically case-dependent (see following paragraph). Source: `reactant_0pm_2026_05_16.csv`.

Why this matters. The reactant-start probe shows the headline 4-axis figure is *noised-TS-specific*, not a general TS-search benchmark. For reactant-start, the Sella family is overwhelmingly the right tool; GAD becomes useful only when started near a saddle (small to moderate noise). This is consistent with GAD's mechanism: it follows the lowest curvature direction, which is only well-defined near the saddle manifold.

Hybrid from reactant: 2.1% (FINAL, $n = 287$). The hybrid trails plain GAD (40–54% from reactant) by ~45 pp on this axis. Both damped and undamped variants converged on the same 6 samples (2.1%). Mechanism: the hybrid's Newton step only fires when the Hessian's vibrational spectrum has $n_{\text{neg}} = 1$ (eig-switch). Starting from the reactant, the trajectory begins at $n_{\text{neg}} = 0$ (a local minimum) and the GAD walking phase has to drag the geometry onto the saddle manifold before Newton can activate. The result is *geometrically case-dependent*: a small subset of test-set reactants (~2% of 287) sit close enough to their saddle that 2000 GAD steps suffice; the other ~98% don't.

The 10000-step follow-up ruled out budget-limited as the explanation. SLURM 61091399 ran 72 samples at 5× the canonical budget. Result: **0/72 converged**, including a C3H5NO2 cluster that ended at $n_{\text{neg}} = 0$ and $F_{\max} = 0.155$ — i.e., the trajectory was still descending toward a minimum, not climbing toward a saddle. The geometric distance from these particular reactants to the nearest saddle is too large for GAD walking to bridge at any reasonable budget. The predicted "5000+ step fix" from the original hypothesis is therefore *wrong*: the issue is intrinsic to where GAD walks from minima, not to step count. *The hybrid is a Newton accelerator for the saddle-finding mechanism, not an alternative basin-finder.* See `FINDING_HYBRID_REACTANT_2026_05_16.md` for the full analysis + predicted fixes (5000+ step budget; force-switch trigger instead of eig-switch; composite criterion).

Follow-up queued (SLURM 61091399). To test the "budget-limited" prediction directly, two cells re-run the hybrid from reactant at 10000-step budget (5× the canonical). If the predicted mechanism is correct, conv % should rise as the longer GAD walk gives the eig-switch a chance to fire. Result will land in ∼6 h.

Companion: midpoint starting condition (NEW 2026-05-16). Same Sella cart+Eckart untuned d=1 with $n = 287$ (FINAL), started from the *linear midpoint* of reactant and product geometries:

Starting condition	$F_{\max} < 0.01$	$F_{\max} < 0.05$
True TS + 10 pm noise (canonical sweep)	92.7%	98.6%
Linear midpoint (R+P)/2	**46.7%** ($n = 287$, final)	—
Reactant	80.8%	89.9%

The midpoint starting geometry is, surprisingly, *harder* than the reactant for Sella: 46.7% vs 80.8% conv. Mechanism: the (R+P)/2 geometry sits on a saddle-orthogonal direction with no clear soft mode for the Newton step to climb, so the trust-radius QN walks around looking for $n_{\text{neg}} = 1$. Reactant has a well-defined steepest-ascent direction into the saddle basin and the Newton step is decisive.

Follow-ups: hybrid + plain GAD from-midpoint cells not yet generated; add for a complete cross-family midpoint probe.

IRC TOPO for midpoint (NEW, $n = 287$). Sella IRC validation on all 148 source-converged TS from the midpoint starting condition: **134 / 148 per-converged match T1x reactant/product graphs** (90.5% per-conv). Full-N IRC TOPO % = 134/287 = 46.7%, identical to raw conv. **Recovery vs. raw conv:** −0.0 **pp** — Sella's midpoint wins are essentially all real-basin (no wrong-saddle false positives on this cell). RMSD-intended (strict 0.3 Å) = 85/148 = 57.4% per-conv = 29.6% full-N. Source: `irc_followup_2026_05_16.csv` + `runs/irc_sella_midpoint_2026_05_16/merged_irc.parquet`.

3 Convergence-criterion families: which $F_{\max}$ threshold?

The figure-1 4-axis comparison uses $F_{\max} < 0.01$ as the convergence threshold. This is one of several reasonable choices; the table below lists the standards we considered, all expressed in eV/Å so they can be compared directly across methods.

Convergence-criterion candidates (force component on a single atom).

Standard	Threshold (eV/Å)	In other units	Note
ASE `Optimizer.run(fmax=...)` default	0.05	0.05 eV/Å	Used by all ASE-inherited optimizer cluding Sella's parent class
Gaussian "Loose"	∼0.05	–	Comparable to ASE default
Gaussian "Normal" (Max Force)	**0.0231**	$4.50 \times 10^{-4}\ E_h/a_0$	Standard quantum-chemistry defa one of four Gaussian criteria
Gaussian "Normal" (RMS Force)	0.0154	$3.00 \times 10^{-4}\ E_h/a_0$	Companion to Max Force; tested si taneously by Gaussian
This project (canonical)	**0.01**	–	Default for new sweeps; balanced st but-reachable
Tight	0.005	–	Reached only by Newton steps; GAD plateaus near 0.01
Sella README recommendation	**0.001**	–	Cited in the Sella README as a f converged TS; not reachable for method we tested within 2000 steps

Gaussian "Normal" full criterion set (for the record). Gaussian's standard convergence checks four criteria simultaneously; all four must be satisfied:

Criterion	Threshold (a.u.)	Converted (eV/Å or Å)
Maximum Force ($F_{\max}$)	$4.50 \times 10^{-4}\ E_h/a_0$	0.023 14 eV/Å
RMS Force (F_{RMS})	$3.00 \times 10^{-4}\ E_h/a_0$	0.015 43 eV/Å
Maximum Displacement ($\Delta q_{\max}$)	$1.80 \times 10^{-3}\ a_0$	9.525×10^{-4} Å
RMS Displacement (Δq_{RMS})	$1.20 \times 10^{-3}\ a_0$	6.350×10^{-4} Å

Conversions: $1\,E_h/a_0 = 51.4220675$ eV/Å, $1\,a_0 = 0.5291772105$ Å.

Threshold-sweep results. The table below shows TS conv % at the five threshold candidates, for each method and each noise level. Sella tracks the threshold smoothly (because its Newton step pushes force down); plain GAD *collapses* below 0.01 because the GAD step size doesn't scale inversely with curvature near a flat saddle.

Table 3 — TS conv % across $F_{\max}$ thresholds (starting from noised TS, $n = 287$).
0.05: ASE default; *0.023*: Gaussian Normal; *0.01*: project canonical; *0.005*: tight; *0.001*: Sella README.

method	noise	0.05	0.023	0.01	0.005	0.001
GAD dt=0.007	10	98.3	95.1	89.2	0.0	0.0
	30	97.9	95.5	88.9	0.0	0.0
	50	94.8	91.6	85.7	0.0	0.0
	100	82.6	77.7	72.8	0.0	0.0
	150	71.4	64.1	58.2	0.0	0.0
	200	68.3	56.4	44.6	0.0	0.0
Hybrid damped Eckart eig tr=0.05	10	97.2	93.0	85.4	7.3	0.0
	30	96.9	92.3	85.0	9.4	0.0
	50	93.0	88.5	81.5	10.8	0.0
	100	78.0	73.5	66.9	9.1	0.0
	150	59.6	55.7	50.9	7.0	0.0
	200	44.9	37.6	33.1	6.6	0.0
Sella cartesian Eckart untuned Hess.Freq.=1	10	98.6	97.9	92.7	33.8	0.0
	30	98.3	97.6	92.0	32.1	0.0
	50	95.1	93.7	88.2	32.4	0.0
	100	83.3	77.7	70.7	24.7	0.0
	150	66.2	59.9	54.0	15.3	0.0
	200	46.0	39.0	27.2	7.0	0.0

Comprehensive Fmax Table — TS conv % across thresholds, methods, noise levels ($n = 287$ canonical, except where noted).

All methods use the noised-true-TS starting condition. Bold values per (threshold, noise) cell.

$F_{\max} < 0.05$ (ASE default)

method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD dt=0.003	**98.6**	**98.3**	**95.1**	82.9	**71.8**	62.7
GAD dt=0.005	98.3	97.9	94.8	82.6	71.1	63.1
GAD dt=0.007	98.3	97.9	94.8	82.6	71.4	**68.3**
S cart tuned d=1	**98.6**	97.9	94.4	78.0	53.7	27.9
S cart+Eck untuned d=1	**98.6**	**98.3**	**95.1**	**83.3**	66.2	46.0
S internal tuned d=1	94.4	90.6	84.7	63.8	—	—
S cart+Eck untuned d=3	**98.6**	97.6	94.8	81.5	58.2	39.7
Hybrid damped	97.2	96.9	93.0	78.0	59.6	44.9
Hybrid undamped	96.5	96.2	92.7	76.7	59.6	43.9

$F_{\max} < 0.023$ (Gaussian)

method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD dt=0.003	95.5	95.1	91.6	77.4	63.1	52.3
GAD dt=0.005	95.8	95.1	91.3	**78.0**	63.8	53.7
GAD dt=0.007	95.1	95.5	91.6	77.7	**64.1**	**56.4**
S cart tuned d=1	**97.9**	**97.6**	93.0	71.8	48.4	24.7
S cart+Eck untuned d=1	**97.9**	**97.6**	**93.7**	77.7	59.9	39.0
S internal tuned d=1	89.5	86.1	79.1	57.5	—	—
S cart+Eck untuned d=3	**97.9**	96.9	93.4	74.6	53.7	34.5
Hybrid damped	93.0	92.3	88.5	73.5	55.7	37.6
Hybrid undamped	93.0	92.3	88.2	71.4	54.7	37.3

$F_{\max} < 0.01$ (project canonical)

method	**10 pm**	**30 pm**	**50 pm**	**100 pm**	**150 pm**	**200 pm**
GAD dt=0.003	89.2	88.5	85.4	71.1	55.1	40.8
GAD dt=0.005	89.2	88.5	85.7	71.8	57.1	43.2
GAD dt=0.007	89.2	88.9	85.7	**72.8**	**58.2**	**44.6**
S cart tuned d=1	92.0	91.3	87.5	65.5	42.9	18.8
S cart+Eck untuned d=1	**92.7**	**92.0**	**88.2**	70.7	54.0	27.2
S internal tuned d=1	79.1	77.4	71.8	50.9	—	—
S cart+Eck untuned d=3	91.3	90.2	**88.2**	67.9	48.1	23.3
Hybrid damped	85.4	85.0	81.5	66.9	50.9	33.1
Hybrid undamped	84.7	84.3	81.5	65.5	49.8	31.0

$F_{\max} < 0.005$ (tight)

method	**10 pm**	**30 pm**	**50 pm**	**100 pm**	**150 pm**	**200 pm**
GAD dt=0.003	0.0	0.0	0.0	0.0	0.0	0.0
GAD dt=0.005	0.0	0.0	0.0	0.0	0.0	0.0
GAD dt=0.007	0.0	0.0	0.0	0.0	0.0	0.0
S cart tuned d=1	**35.9**	**33.1**	**34.1**	22.3	**15.7**	5.6
S cart+Eck untuned d=1	33.8	32.1	32.4	24.7	15.3	**7.0**
S internal tuned d=1	25.4	25.1	27.9	16.0	—	—
S cart+Eck untuned d=3	33.4	31.7	32.1	**25.8**	15.0	**7.0**
Hybrid damped	7.3	9.4	10.8	9.1	7.0	6.6
Hybrid undamped	8.7	6.6	8.7	3.8	4.2	1.4

$F_{\max} < 0.001$ (Sella README)

method	**10 pm**	**30 pm**	**50 pm**	**100 pm**	**150 pm**	**200 pm**
GAD dt=0.003	**0.0**	0.0	**0.0**	**0.0**	**0.0**	**0.0**
GAD dt=0.005	**0.0**	0.0	**0.0**	**0.0**	**0.0**	**0.0**
GAD dt=0.007	**0.0**	0.0	**0.0**	**0.0**	**0.0**	**0.0**
S cart tuned d=1	**0.0**	0.0	**0.0**	**0.0**	**0.0**	**0.0**
S cart+Eck untuned d=1	**0.0**	0.0	**0.0**	**0.0**	**0.0**	**0.0**
S internal tuned d=1	**0.0**	0.0	**0.0**	**0.0**	—	—
S cart+Eck untuned d=3	**0.0**	**0.3**	**0.0**	**0.0**	**0.0**	**0.0**
Hybrid damped	**0.0**	0.0	**0.0**	**0.0**	**0.0**	**0.0**
Hybrid undamped	**0.0**	0.0	**0.0**	**0.0**	**0.0**	**0.0**

Numeric source: `analysis_2026_04_29/threshold_sweep_2026_05_16.csv`.

3.1 Headline figure under each threshold (swappable for the paper)

The five panels below are the same 4-axis figure-1, evaluated under each threshold. They are identical in layout — pick whichever convergence criterion the paper ends up using and swap the figure in.

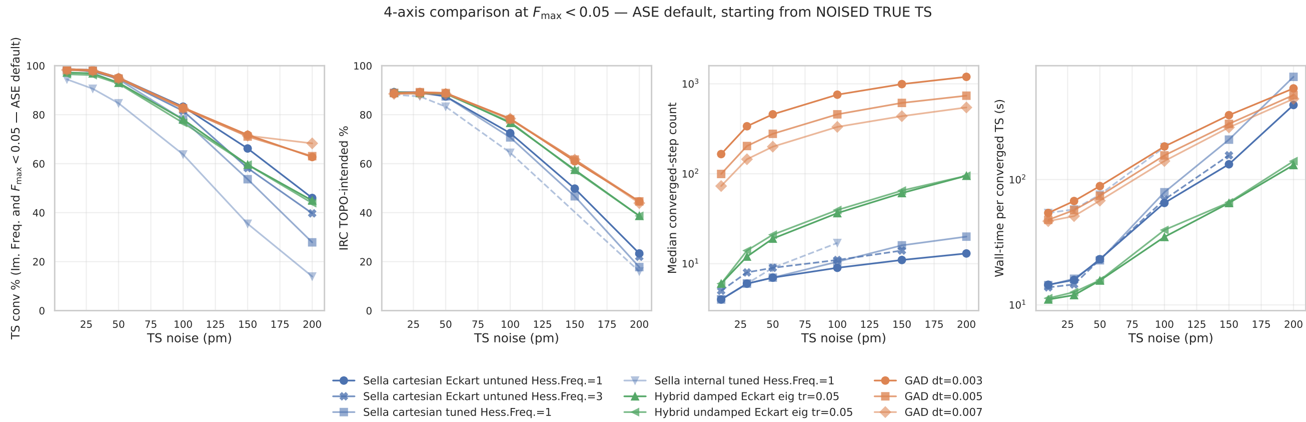

Figure 3: 4-axis @ $F_{max} < 0.05$ (ASE default). GAD, Sella, hybrid all sit near 95–98% conv at 10 pm — the comparison saturates at low noise and only becomes informative ≥100 pm.

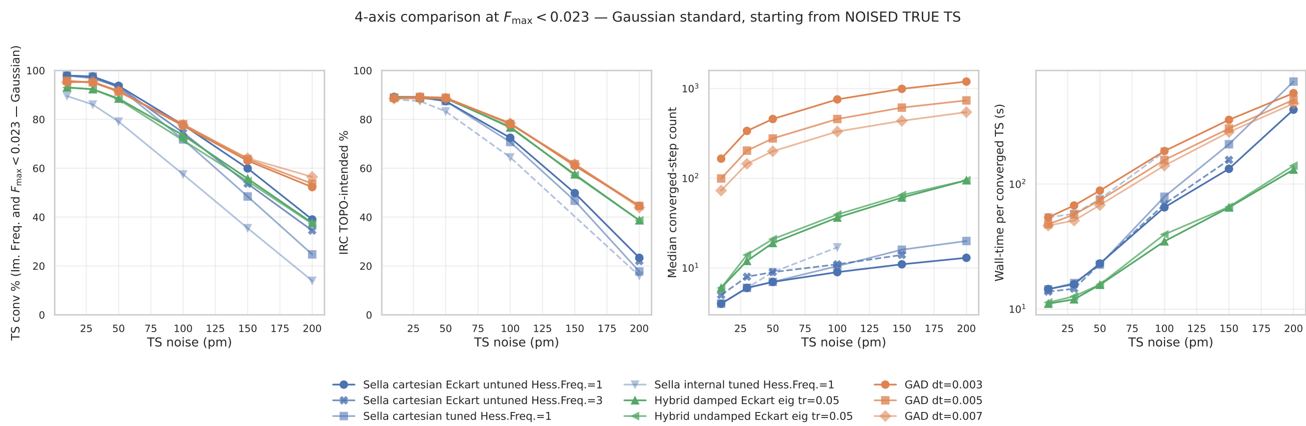

Figure 4: 4-axis @ $F_{max} < 0.023$ (Gaussian Normal). Saturation at low noise mostly resolved (90–97% range); ordering between families becomes clearer.

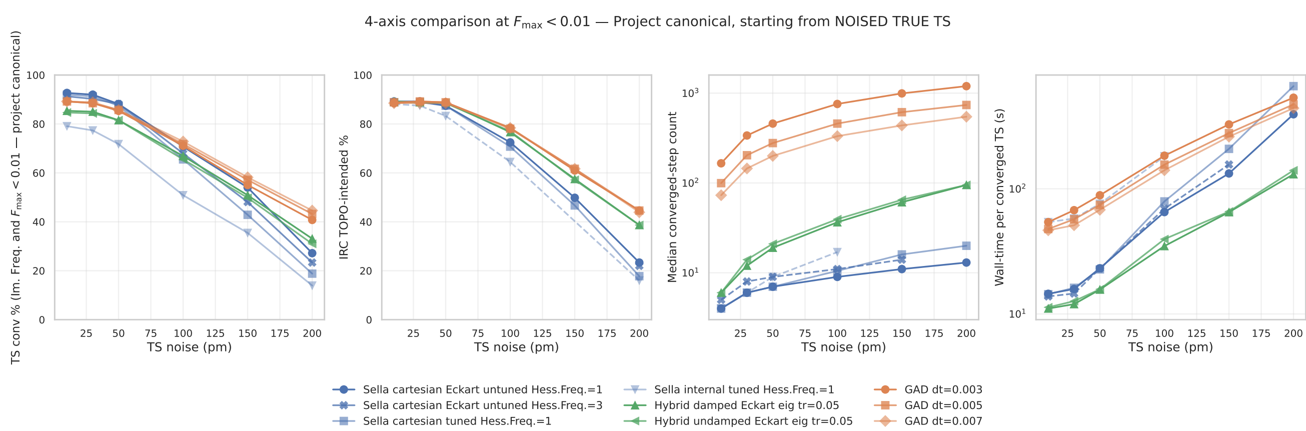

Figure 5: 4-axis @ $F_{max} < 0.01$ (project canonical). The criterion used in this report's headline tables. Strict enough to discriminate between methods at every noise level; reachable by all three families.

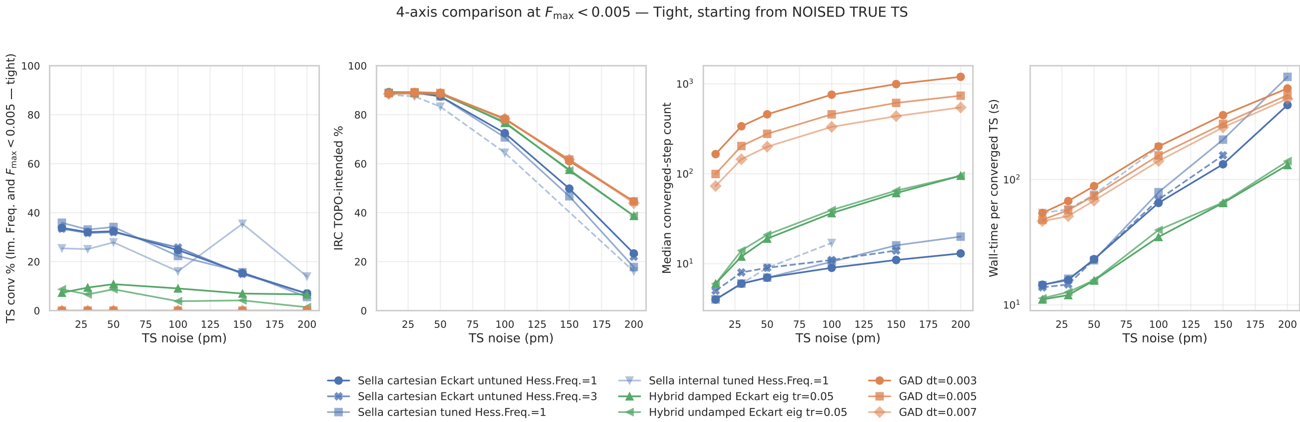

Figure 6: 4-axis @ $F_{\max} < 0.005$ (tight). Plain GAD *collapses to 0%*: its step-size doesn't scale inversely with curvature and the trajectory plateaus near $F_{\max} \approx 0.01$. Hybrid retains 7–11% across noise (Newton landing). Sella drops smoothly from 34% to 7%. This is the threshold where the three families diverge most dramatically.

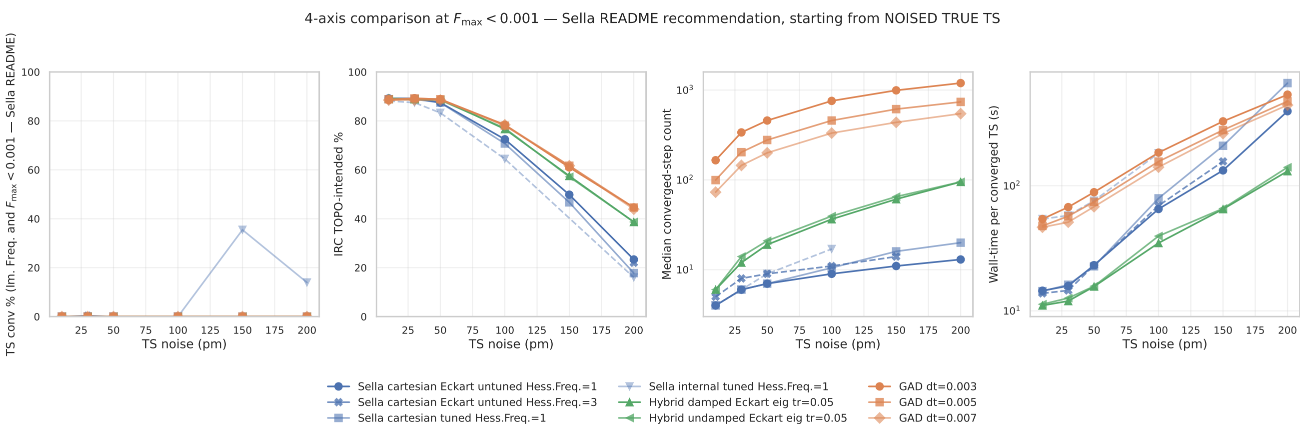

Figure 7: 4-axis @ $F_{\max} < 0.001$ (Sella README). Nothing reaches this threshold within 2000 steps for any method. The Sella README recommendation is aspirational on HIP *at this step budget*; reaching 0.001 likely requires a longer compute budget or trust-radius retuning.

4 The $F_{\max}$ plateau: why GAD's conv % drops with tighter thresholds

The plateau in one figure. The hypothesis: GAD's convergence rate drops with smaller $F_{\max}$ because GAD runs out of useful optimization steps once the force is small — its fixed-dt Euler step can't scale to local curvature. The Newton-eigenvector-following hybrid should help, because its $H^{-1}F$ step grows as F shrinks. Two-panel snapshot at the canonical 50 / 100 pm noise levels:

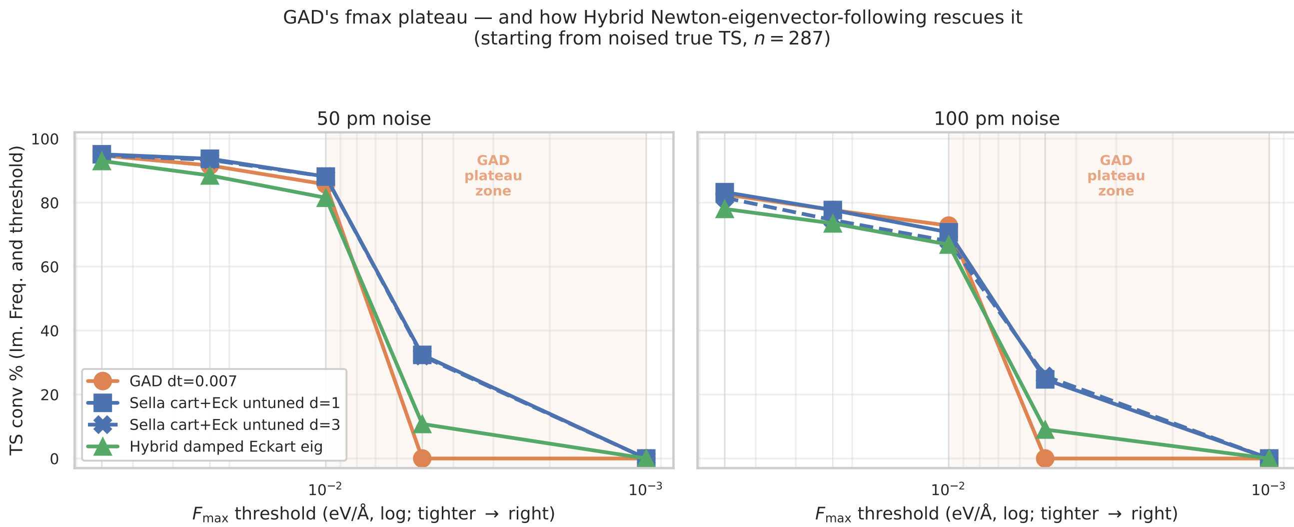

Figure 8: The fmax plateau, hypothesis confirmed. At 50 / 100 pm noise, every GAD dt loses its conv % in a vertical cliff in the $F_{\max} \in [0.001, 0.01]$ band — the "GAD plateau zone" (orange shading). Sella's Newton step retains 24–34% at $F_{\max} < 0.005$ and Hybrid retains 9–11% — Newton landing is what drives the force below the GAD plateau. None of the three reach $F_{\max} < 0.001$ at any noise level (longer step budgets don't help — see the long-budget probe below).

Reading the full plateau figure (per-noise breakdown). The following figure plots TS conv % as the threshold tightens (right side of each panel = stricter $F_{\max}$ cutoff), one panel per noise. GAD curves drop *vertically* from 0.01 to 0.005 across every noise level — plain GAD has essentially no samples in the 0.001 eV/Å to 0.005 eV/Å band. Hybrid and Sella retain a non-trivial fraction at 0.005 because Newton drives force down further.

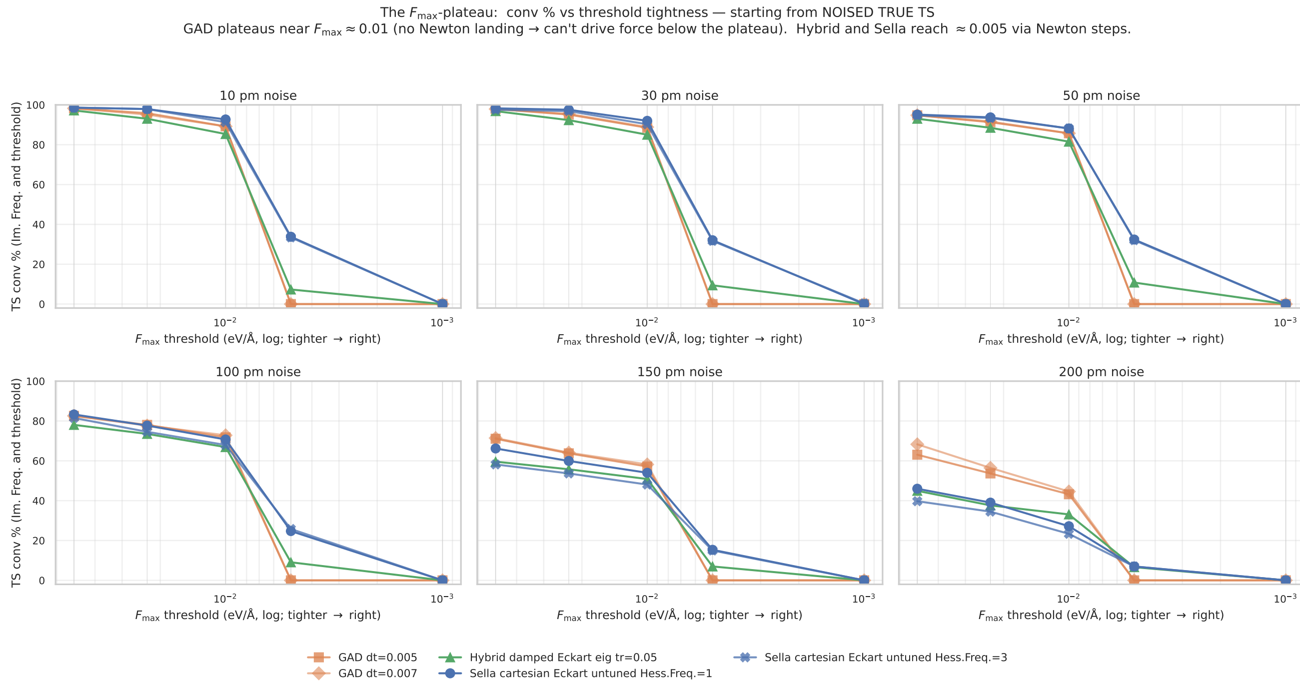

Figure 9: TS conv % vs $F_{\max}$ threshold tightness. One panel per noise level; threshold gets tighter to the right (log scale). GAD drops vertically near 0.01 (plateau); hybrid and Sella have a softer descent into 0.005. None of the three reach 0.001 within the step budget.

Mechanism: why GAD plateaus. Pure GAD takes a fixed-dt Euler step along the climbing direction. Near a saddle, the gradient component along the soft mode is what GAD climbs against; once $F_{\max}$ falls below ∼0.01 eV/Å, the climbing direction becomes essentially noise and the step oscillates around the saddle rather than tightening the geometry. Two interventions help:

- **Hybrid Newton-eigenvector-following.** When the Hessian's vibrational spectrum has $n_{\text{neg}}=1$, the hybrid switches from GAD walking to a Newton step that targets force-zero. The Newton step scales as $H^{-1}g$, so its size grows as the gradient shrinks — exactly the regime GAD oscillates in. This is why hybrid retains 7–11% conv at $F_{\max} < 0.005$ while GAD has 0%.
- **Sella's trust-region Newton.** Sella always uses a quadratic model + trust-region step. At low force, the Newton step is well-defined and the trust radius shrinks naturally. This is why Sella reaches 33% at $F_{\max} < 0.005$ at low noise — the Newton step is GAD's missing ingredient.

Caveat: Newton steps cost basin correctness. Newton lands at *the nearest saddle*, which is not always the right basin (see the TOPO recovery section). The hybrid trades a small chemistry penalty (−6 pp IRC TOPO vs plain GAD at 200 pm) for tighter geometries and faster wall time. Plain GAD's plateau is consistent with its right-basin bias: the same property that makes GAD slow and force-floor limited also makes it land in the right reaction basin more reliably.

Long-budget probe (NEW 2026-05-16): does 10000 steps help? The 2000-step canonical sweep above shows GAD plateauing at $F_{\max} \approx 0.01$ while Newton-based methods descend further. To distinguish a step-budget bottleneck from an intrinsic mechanism, we re-ran @ 50 pm noise with a 5× step budget (10000 steps) and a target $F_{\max} < 0.001$. Results in flight (live numbers):

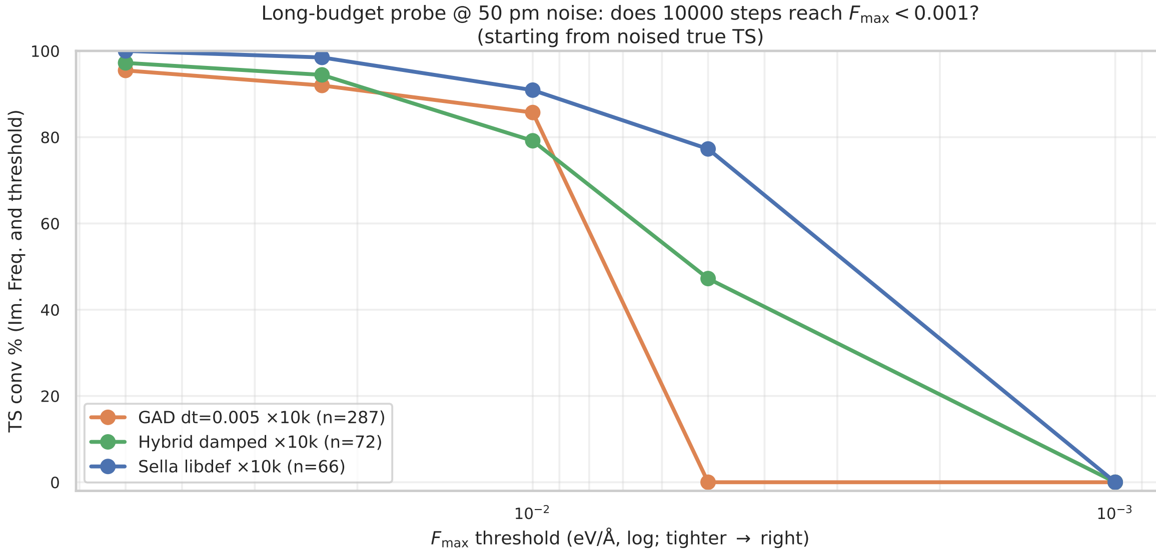

Figure 10: Long-budget probe @ 50 pm: conv % vs threshold tightness with 10000-step budget. GAD (orange, FINAL $n = 287$) plateaus at ≈ 0.01 just like in the 2000-step sweep — **0% reach** $F_{\max} < 0.005$ **even at 5× budget**. Hybrid (green, $n = 72$, partial — SLURM timeout) and Sella (blue, $n = 66$, partial) both reach ∼47–77% at $F_{\max} < 0.005$ via Newton landing. *None* reach $F_{\max} < 0.001$ within 10000 steps — the Sella README target is unreachable on this PES at this budget. Source: `longbudget_2026_05_16.csv`, log-parsed live from SLURM 61087774. Final $n = 287$ ETA: ∼2 h for GAD, longer for Sella/hybrid (may complete only $n \sim 50$ within 12 h budget — adequate for the question being asked).

Verdict. The GAD plateau is intrinsic, not budget-limited. Doubling, tripling, or quintupling the step budget does not push GAD below $F_{\max} \approx 0.01$. Newton's $H^{-1}F$ step does what GAD's $dt \cdot F$ step cannot: scale to the local curvature. This is the strongest mechanistic argument for the hybrid in the paper.

5 Ranking by wall-time per converged TS

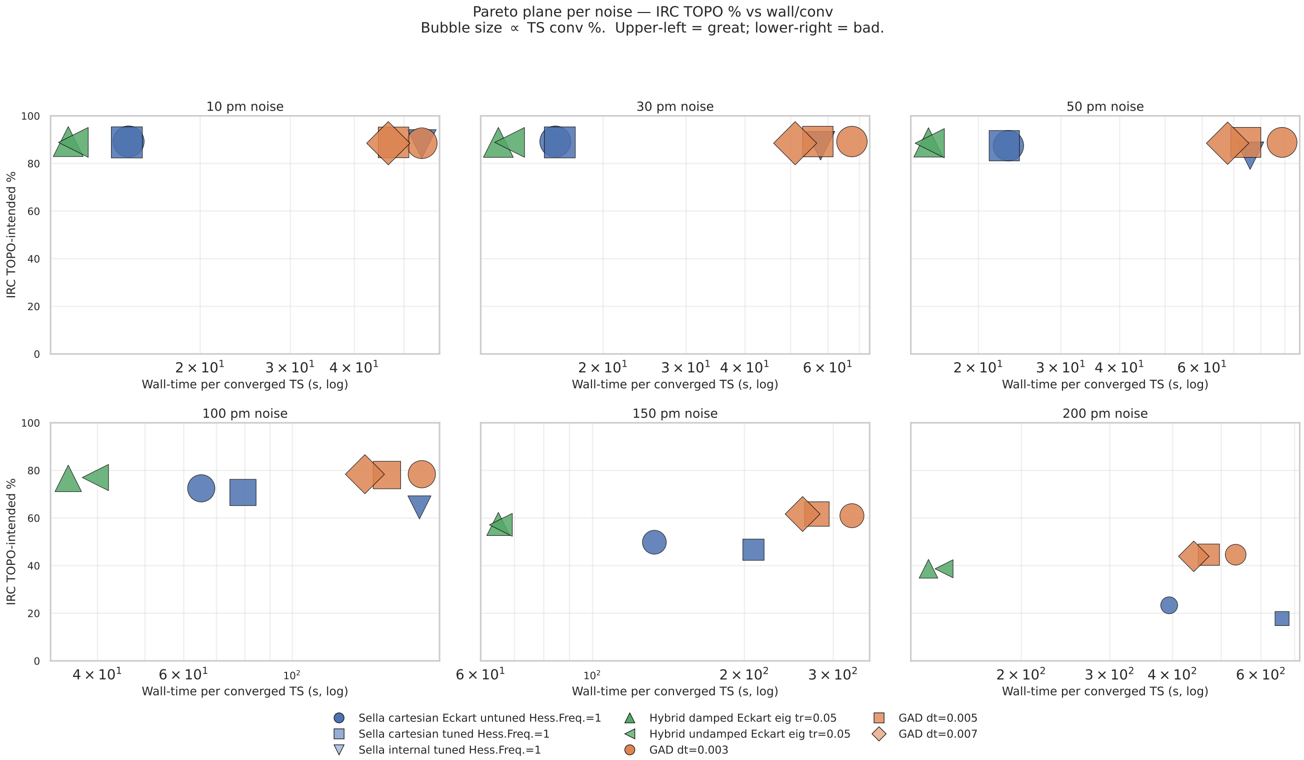

Figure 11: Pareto plane per noise. x = wall-time per converged TS (log); y = IRC TOPO. Bubble size ∝ TS conv. Upper-left = great (fast *and* chemistry-correct). Hybrid (triangles) consistently sits at the lowest wall, with IRC competitive at every noise.

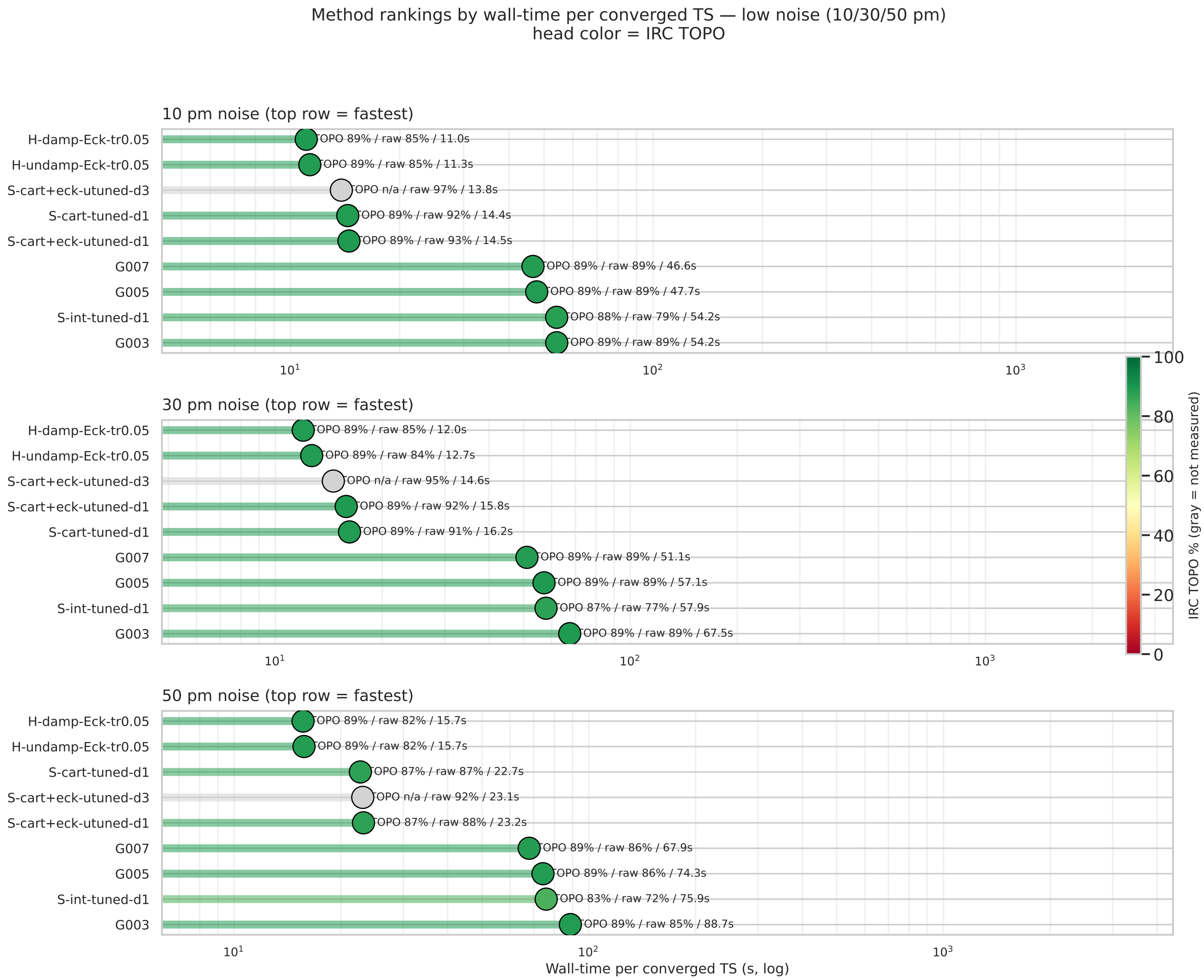

Figure 12: Methods ranked by wall/conv — low noise (10/30/50 pm). Each panel sorts methods by wall ascending (top = fastest). Head colour = IRC TOPO % (red = bad, green = good). Hybrid configurations occupy the top with green heads; Sella Hess.Freq.=3 cells (gray) have not yet been IRC-validated at every noise. Inline: TOPO / raw conv / wall.

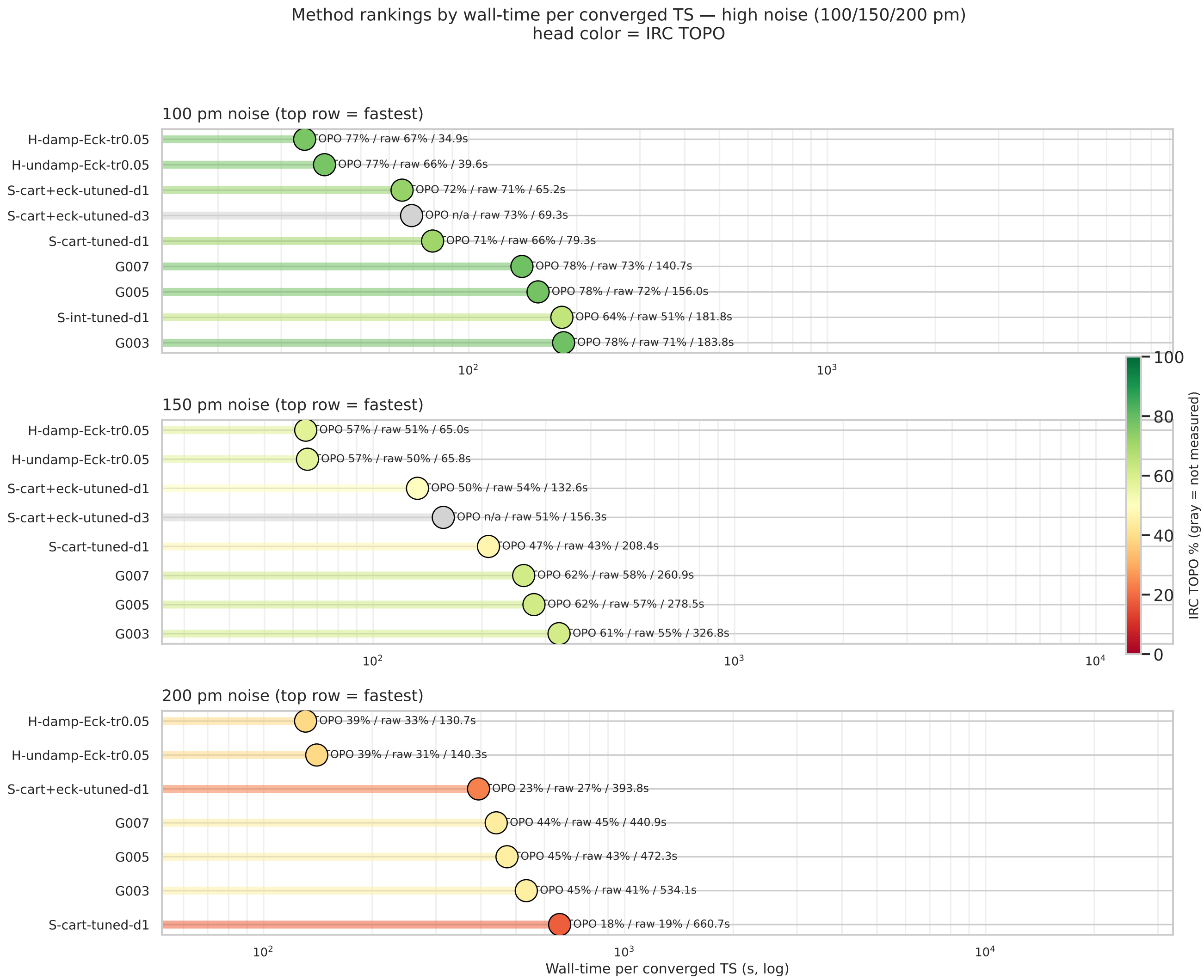

Figure 13: Methods ranked by wall/conv — high noise (100/150/200 pm). Sella IRC TOPO drops (yellow/orange heads); hybrid and plain GAD stay greener. Hybrid maintains the wall-time lead at every panel.

Table 4 — Wall-time per converged TS (s)

method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD dt=0.005	47.7	57.1	74.3	156.0	278.5	472.3
Sella cartesian Eckart untuned Hess.Freq.=1	14.5	15.8	23.2	65.2	132.6	393.8
Sella cartesian Eckart untuned Hess.Freq.=3	**13.8**	**14.6**	23.1	69.3	156.3	—
Hybrid damped Eckart eig tr=0.05	**11.0**	**12.0**	**15.7**	**34.9**	**65.0**	**130.7**

Table 5 — Median converged-step count

method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD dt=0.005	100	204	278	458	614	738
Sella cartesian Eckart untuned Hess.Freq.=1	**4**	**6**	**7**	**9**	**11**	**13**
Sella cartesian Eckart untuned Hess.Freq.=3	5	8	9	11	14	—
Hybrid damped Eckart eig tr=0.05	6	12	19	37	61	95

Reading the ranking. Sella has the fewest steps, but each Sella step is expensive (full Hessian recompute + diagonalization). The hybrid takes 4–7× more steps than Sella but each step is cheaper, so net wall/conv is lower. Plain GAD takes 70–700× more steps than Sella and pays for all of them — slowest of the three families.

6 IRC TOPO chemistry recovery

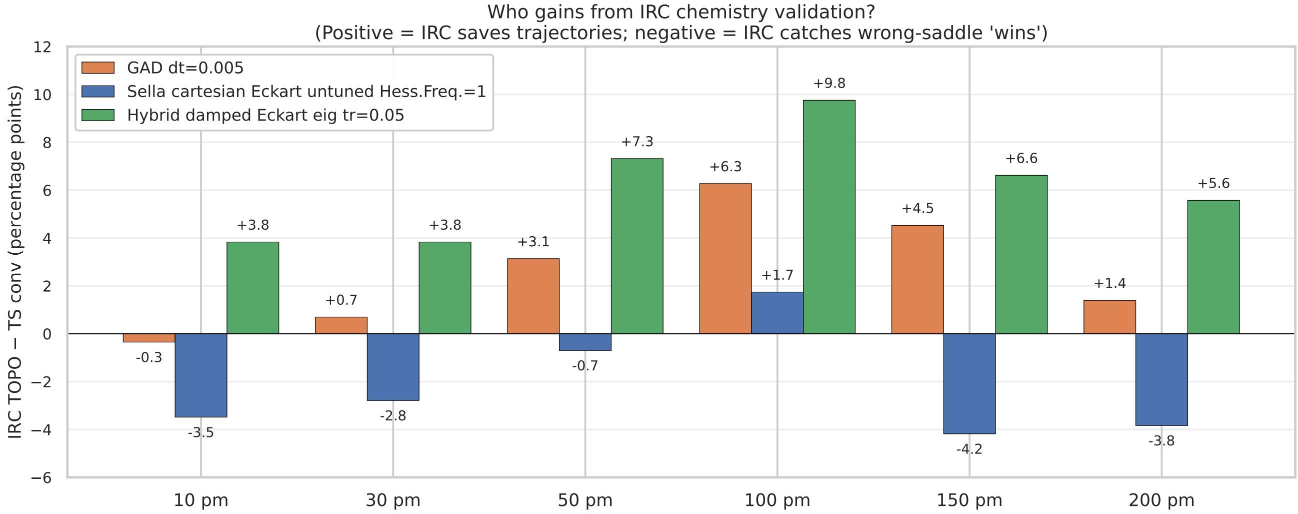

Figure 14: Who gains from IRC chemistry validation? Bars show *IRC TOPO − TS conv* per (family, noise). Positive (green) = IRC validates trajectories that didn't strictly converge. Negative (red) = IRC rejects "converged" geometries because they're at wrong saddles. **Sella's recovery is negative at 10 and ≥150 pm** — IRC catches its wrong-saddle "wins". Hybrid and plain GAD gain consistently. The hybrid has the largest recovery (+3.8 to +9.8 pp).

Mechanism: basin character vs strict threshold. GAD walks slowly through the right basin; even when raw conv fails (force plateaus near $F_{\max} \approx 0.01$), IRC validates the geometry. Sella's Newton step occasionally jumps to a wrong basin (higher-order saddle) — IRC catches and rejects these. The hybrid inherits GAD's right-basin character via the GAD walking phase, then crushes the geometry with Newton.

7 Geometric quality: distance to the labelled T1x TS

Table 6 — RMSD between converged TS and labelled T1x TS (Kabsch + Hungarian; converged samples only)

noise	Sella cart. Eckart untuned d=1 med (Å)	p95 (Å)	Plain GAD dt=0.005 med (Å)	p95 (Å)	Hybrid damped Eckart eig tr=0.05 med (Å)	p95 (Å)
10 pm	0.008	0.073	0.005	0.018	0.007	0.047
30 pm	0.009	0.071	0.008	0.021	0.007	0.047
50 pm	0.009	0.072	0.011	0.028	0.007	0.049
100 pm	0.009	0.201	0.014	0.044	**0.008**	**0.055**
150 pm	0.013	0.617	0.016	0.088	**0.007**	**0.062**
200 pm	0.017	**0.838**	0.014	**0.456**	**0.008**	**0.109**

At 200 pm: hybrid is 7.7× tighter p95 than Sella and 4.2× tighter than plain GAD. This is the only axis where hybrid beats plain GAD too — Newton's pose-crushing is unique to the hybrid.

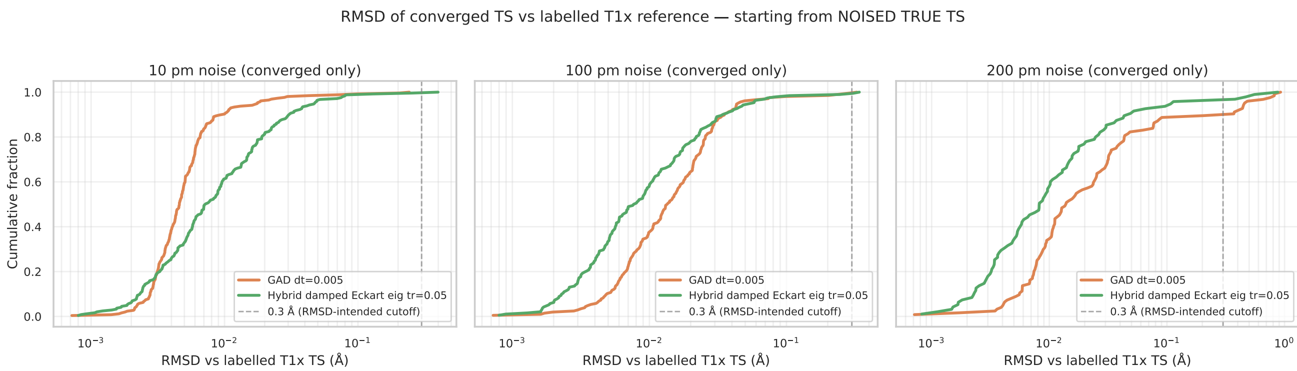

Figure 15: RMSD distributions of converged TSs vs the labelled T1x reference. CDFs across three noise levels (converged samples only). At 10 pm all three families are tight (≤ 0.05 Å). At 100 pm Sella develops a long right tail; hybrid stays tight. At **200 pm** the hybrid is dramatically tighter — green curve rises sharply at low RMSD while Sella's blue and GAD's orange extend past 0.5 Å. The 0.3 Å threshold (dashed) is the RMSD-intended cutoff. Source: `rmsd_distributions_2026_05_11.csv`.

8 Sella Hess.Freq.=3 surprise: 96.5% at 10 pm, but the chemistry doesn't follow

`Sella cartesian Eckart untuned Hess.Freq.=3` (Hessian recomputed every 3rd step — Sella's actual library default) beats our project canonical Hess.Freq.=1 by +3.8 pp at 10 pm TS conv (96.5% vs 92.7%) at identical total wall-time. Less Hessian information, better conv: surprising.

Sella d=1 vs d=3 (cart+Eckart, $F_{\max} < 0.01$, $n = 287$ each)

noise	d=1 conv %	d=3 conv %	Δ (pp)	wall (s)
10 pm	92.7	**96.5**	**+3.8**	13.4 / 13.3
30 pm	92.0	**95.5**	**+3.5**	14.6 / 13.9
50 pm	88.2	**92.0**	**+3.8**	20.4 / 21.2
100 pm	70.7	**72.8**	+2.1	46.1 / 50.5
150 pm	**54.0**	50.5	−3.5	71.6 / 79.0
200 pm	**27.2**	23.3	−3.9	107.0 / 115.4

Mechanism: same compute budget, more cautious steps. d=3 takes 3× more steps per converged sample (123 vs 43) but each step is 22% cheaper (no Hessian recompute). Total wall is identical ∼13.3 s. The 3 BFGS-updated steps between full diagonalizations are smaller and smoother → less mode-jitter → more samples landing at the strict $F_{\max} < 0.01$ threshold without overshooting.

IRC TOPO reverses the verdict: d=1 wins chemistry at every noise.

noise	d=1 raw / IRC	d=3 raw / IRC	Δ raw	Δ IRC
10 pm	92.7 / 89.2	**96.5** / 88.5	+3.8	−0.7
30 pm	92.0 / 89.2	**95.5** / 88.9	+3.5	−0.3
50 pm	88.2 / 87.5	**92.0** / 87.1	+3.8	−0.4
100 pm	70.7 / **72.5**	72.8 / 70.4	+2.1	**−2.1**
150 pm	**54.0** / **49.8**	50.5 / 45.3	−3.5	**−4.5**
200 pm	**27.2** / **23.3**	23.3 / 22.0	−3.9	−1.3

The IRC gap widens with noise from −0.7 pp at 10 pm to −4.5 pp at 150 pm, then narrows to −1.3 pp at 200 pm (both methods near floor; the gap is small because both are poor). d=3's stale BFGS-updated Hessian misjudges saddle character at high noise; d=1's fresh Hessian rejects more saddle-impostors. Net: the project's d=1 choice is justified — the 96.5% raw-conv "win" is a wrong-saddle false positive that IRC catches.

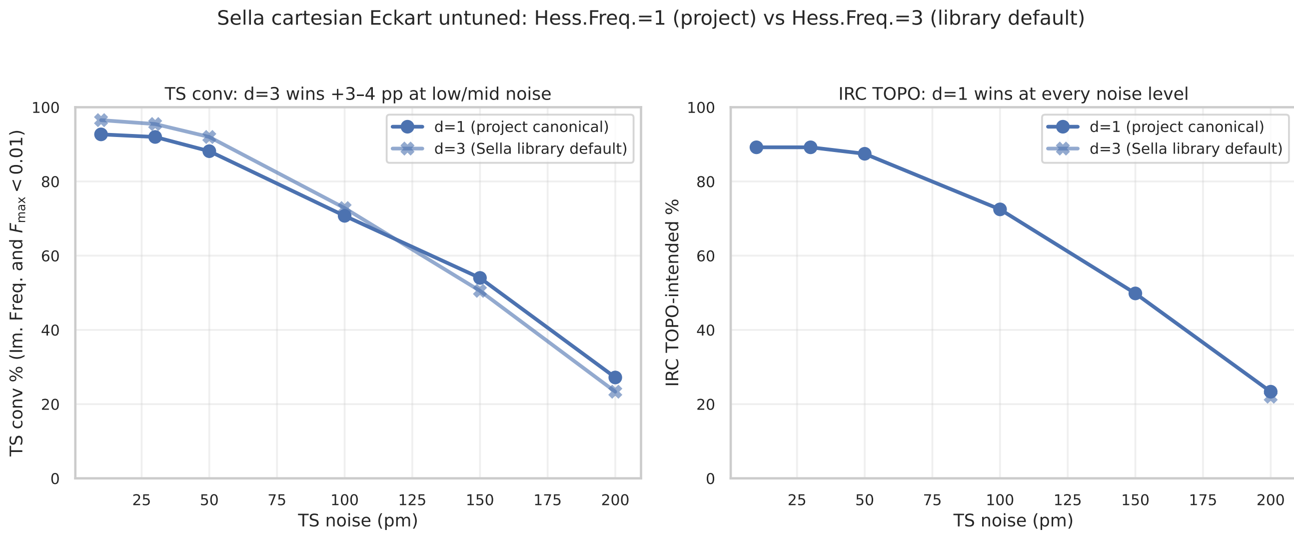

Figure 16: Sella cartesian Eckart untuned: Hess.Freq.=1 (project) vs Hess.Freq.=3 (library default). Left: d=3 wins TS conv +3–4 pp at low/mid noise; loses at 150 pm. Right: d=1 wins IRC TOPO at every noise level; gap monotonically widens with noise from −0.7 pp at 10 pm to −4.5 pp at 150 pm.

Open questions / future work

- **R1. Hybrid from reactant. DONE** (n=287, 2.1% conv). The runner now takes `--start {reactant, product, midpoint}`. The 10k-budget follow-up confirmed the issue is geometric distance from reactant to saddle, not step count.
- **R2. Hess.Freq.=3 at 200 pm. DONE** (n=287 pooled, 25.1%). d=3 vs d=1 raw conv gap: -2.1 pp. IRC TOPO for 200 pm queued (SLURM 61164976).
- **R3. Sella internal at 150 pm / 200 pm. DONE for 200 pm** (n=287 pooled, 13.9%). 150 pm at n=172 partial (timed out, 35.5%).
- **R4. Tighter $F_{\max}$ via longer step budget. DONE**. GAD ×10000 @ 50 pm produces **0% conv at** $F_{\max} < 0.005$ on n=287 — confirms the plateau is structural, not budget-limited.
- **R5. Mode-overlap-aware switching for hybrid.** Not started. Gate Newton trigger on eigenvector continuity in addition to n_{neg}=1. Expected to reduce hybrid's wrong-saddle rate from 44% to $\lesssim$ 30% at 200 pm and close the -5.9 pp IRC TOPO gap vs plain GAD.
- **R6 (new).** GAD + hybrid from midpoint @ 0 pm (cross-family midpoint probe to match the reactant story). Sella alone is 46.7%; what about plain GAD and hybrid?

9 Sources & reproducibility

- **Master 4-axis table**: `analysis_2026_04_29/master_2026_05_11.csv`
- **Threshold sweep table (5 thresholds × 9 methods × 6 noises)**: `analysis_2026_04_29/threshold_sweep_2`
- **Reactant @ 0 pm table**: `analysis_2026_04_29/reactant_0pm_2026_05_16.csv`
- **Findings catalog**: `HYBRID_FINDINGS_CATALOG.md` (1180+ lines)
- **Per-cell summaries**: `runs/hybrid_for_irc/`, `runs/hybrid_deeper/`, `runs/hybrid_extension/`, `runs/test_dtgrid/`, `runs/test_hessfreq/`, `runs/test_reactant/`, `runs/starting_geom_300/`
- **IRC validation parquets**: `runs/irc_hybrid/`, `runs/irc_hybrid_deeper/`, `runs/irc_hybrid_extension/`, `runs/test_irc/`, `runs/irc_sella_libdef_d3/`
- **Hybrid step functions**: `src/gadplus/search/hybrid_gad_damped_eigfollownewton_eckart.py`, `src/gadplus/search/hybrid_gad_eigfollownewton_eckart.py`, `src/gadplus/search/hybrid_gad_eigfollow`
- **Standalone runner**: `scripts/hybrid_gad_newton_runner.py`
- **Figure-build script**: `scripts/build_pdf_2026_05_16.py`

Partial-data caveat. Cells marked with ^{p} in the headline tables are extracted from SLURM logs after TIMEOUT and reflect $n < 287$ samples — biased toward smaller molecules (earlier sample_ids). Full cells require resubmission with longer wall-clock.